David Koyrakh

Term Project: Milestone 1

College of Science and Technology, Bellevue University

DSC-540: Data Preparation

Professor Williams

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Subject Area

The aim of this project is to process and visualize demographic data from all 50 states. Socioeconomic variables to be assessed include financial status, employment, marital status, family size, religiosity, and more.

Data Sources

Flat File

For the flat file, I'll be utilizing the USDA Economic Research Service dataset, which provides median household income and unemployment rates throughout the United States for the years 2021 and 2022, respectively. This dataset is a 17-megabyte CSV file. It contains numerous entries for each County and State.

Link:
https://ers.usda.gov/sites/default/files/_laserfiche/DataFiles/48747/Unemployment.csv?v=37077

Website

For the website data source, I plan to utilize a specific table in the Wikipedia article on religiosity across the United States. The table displays data from a 2014 Pew Research study and includes state-by-state metrics for belief in G-d, importance of religion, and frequency of prayer.

Link:
https://en.wikipedia.org/wiki/List_of_U.S._states_and_territories_by_religiosity#cite_ref-61

API

For data sourced from an API, I'll be utilizing the American Community Survey 5-Year Data (2009-2023) from the US Census API, which provides detailed social and financial data throughout the United States.

Link: https://www.census.gov/data/developers/data-sets/acs-5year.html

Relationships

All three data sources are related by state. The USDA CSV contains various employment and income statistics for each state, with a many-to-one relationship to the Pew Research religiosity table. The Census API returns multiple demographic data points for each state, featuring a many-to-many relationship with the USDA CSV, and a many-to-one relationship with the Pew Research table.

Plan

The project will begin with data collection and cleaning from my three sources. For the USDA dataset, I will focus on filtering and aggregating the state-level data, removing county-level entries and checking for missing values. The religiosity data table from Wikipedia requires web scraping in order to extract its contents, followed by data cleaning to ensure that state identifiers, numerical values, and column headers visually match how the table is rendered by Google Chrome. Before retrieving data from the Census API, I'll conduct more research to check if extra parameterization of the API queries is necessary in order to bring the total returned dataset to a reasonable size (perhaps no more than a few hundred megabytes). I will also check for rate limits, which will be respected via explicit delays (and request batching) in the code, and whether the API returns data via pagination. The code will be written in Python. If necessary, asynchronous techniques and libraries (e.g. *asyncio*) will be used to optimize the network requests while respecting rate limits, if any.

The next phase involves data transformation and integration. I will be using capitalized 2-letter state abbreviations as state identifiers for the final data of this project. This field, in particular, is important, as it will be used for relating each dataset together. Therefore, I will need to store the state identifier values from each source in a way that strictly matches the convention described above. The various socioeconomic variable names, such as “median household income” or “household size” will be stored as strings, and I will conduct more research to see if any of these variables are shared across the data sets. Socioeconomic variable types will be handled on a case-by-case basis, since some are strings or numeric, others are Boolean, and there may be None or NaN types for missing values.

The final phase will focus on merging and visualization, as outlined in the instructions for Milestone 5. Some potential challenges in my planned approach and selection of datasets include: handling missing data, respecting the API rate limit, and ensuring accurate state-level aggregations. I am also concerned about the years/timeframes from which the various datasets were collected, since they might not all match or overlap. If they do not overlap at all, that could significantly hurt the value of comparing the datasets and visualizing them within the same study. Usually, socioeconomic data is studied over preselected time frames, since global factors (and time itself) can profoundly influence data. I will do my best to present the data accurately and transparently, including disclaimers, if needed, and stating assumptions made throughout the process.